Vishnu vardhan Madasi

Denton, TX | vishnuvardhan.wf438@gmail.com | 940-629-4683 | LinkedIn

Professional Summary

AI Engineer with hands-on experience designing, building, and deploying production-grade machine learning and Generative AI solutions for enterprise use cases. Proficient in Python, PyTorch, TensorFlow, Scikit-learn, LangChain, and Hugging Face Transformers. Experienced in building LLM-powered applications with RAG pipelines, prompt engineering frameworks, and multi-agent orchestration logic. Skilled in MLOps practices including CI/CD-driven model lifecycle management, model monitoring, evaluation pipelines, and containerized deployment. Proven ability to translate complex business problems into scalable AI-driven solutions and communicate effectively across engineering, product, and stakeholder teams.

Experience

Ez4 Technologies | *AI Engineer* | *Remote* **Jan 2025 – Mar 2026**

- Cut incident response time for the reliability team by building a LangGraph multi-agent system that automatically triaged issues and routed them to the right workflow, replacing a manual process built on searching through 80K+ internal documents.
- Reduced inaccurate AI answers by combining vector search with keyword search (BM25) instead of embeddings alone, improving response accuracy while keeping answers under 5 seconds for use during live incidents.
- Eliminated nightly batch re-indexing by building an incremental ingestion and embedding pipeline that only processed new or changed documents, bringing knowledge-base updates to near real-time.
- Replaced ad-hoc quality checks with evaluation pipelines (Langfuse, LangSmith) built on curated test sets, catching regressions before production and turning "is this better or worse" into a data-backed decision.
- Made the system maintainable beyond a single owner by documenting architecture and design decisions, allowing other engineers to extend the workflows without starting from scratch.
- Extended the same multi-agent framework to additional enterprise use cases beyond the original reliability workflow, multiplying the return on the initial engineering investment.
- Benchmarked throughput, memory, and inference latency across different compute environments to identify the most cost-effective infrastructure that still met response-time requirements.
- Consolidated structured and unstructured data from multiple enterprise sources (Snowflake, Databricks) into unified pipelines, eliminating duplicated data-prep work across AI and analytics teams.
- Built and productionized classification models (LightGBM, XGBoost, CatBoost) for structured business problems, automating retraining and redeployment through MLflow and Airflow to remove manual intervention from the ML lifecycle.

Solware IT Infotech | *AI Engineer* | *Hyderabad, India* **Jul 2022 – Jul 2024**

- Replaced a slow, manual clinical data process with an LLM-powered pipeline, re-engineering the underlying schemas so structured and unstructured patient data could be used directly for inference.
- Gave leadership visibility into AI cost and performance by connecting Langfuse to a BI dashboard tracking cost, latency, and quality, making retraining decisions data-driven instead of anecdotal.
- Automated the clinical data transformation step with REST APIs, removing a significant amount of manual data-wrangling work from the team's workflow.
- Improved audit and compliance readiness by adding versioning (SBAR) and traceability to model outputs, giving the team a clear answer to "why did the model say this" in a regulated clinical setting.
- Reduced downstream errors caused by inconsistent terminology by standardizing medical terms across model outputs.
- Investigated ONNX export and hardware-aware optimization techniques, laying the groundwork for faster, lower-cost model deployment.
- Bridged the gap between stakeholder requests and technical implementation by translating business asks into concrete system architectures for enterprise AI applications.
- Automated a previously manual inspection process by building computer vision tooling (OpenCV, YOLO), deployed at scale on Azure AI Services and Azure Machine Learning.
- Automated dispute and chargeback handling by building a document-extraction pipeline that converted unstructured PDFs and images into structured data, removing manual document review from the process.

Cantilever Labs | *AI/ML Trainee* | *Hyderabad, India* **Feb 2022 – May 2022**

- Automated feature engineering and preprocessing for structured datasets in Python, streamlining data preparation ahead of model training.
- Cut model deployment time by 70% by replacing a manual, multi-step release process with an automated CI/CD retraining pipeline.
- Caught model accuracy and drift issues before they affected users by building inference services and monitoring dashboards tracking KPIs in production.
- Worked with cross-functional teams to ensure ML prototypes solved the actual business problem stakeholders cared about, not just a technically interesting one.

Technical Skills

ML/AI Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CatBoost, Hugging Face Transformers, LangChain, LangGraph, Apache Spark.

Generative AI LLMs: Prompt Engineering, RAG Pipelines, Multi-Agent Orchestration, Vector Databases (FAISS), BM25, LLM Evaluation, Agentic AI Systems, Scalable AI Solutions.

Data Engineering: ETL Workflows, Feature Engineering, Structured Unstructured Data Transformation, REST APIs, Automated Data Pipelines, Data Analytics.

MLOps Deployment: CI/CD Pipelines, Model Versioning, Model Monitoring Logging, Containerization, Langfuse, LangSmith, Heroku, AWS, GCP, Kubeflow, Airflow, MLflow.

Tools Platforms: Git/GitHub, Flask, Linux, OpenCV, Stable Diffusion, CLIP, Matplotlib, Seaborn, Azure AI Services, Azure Machine Learning, Azure Functions, Azure DevOps, GitHub Actions, Snowflake, Databricks, NoSQL Databases, YOLO, PDF/Image Extraction, Claude Code, Cursor.

Education

University of North Texas — Denton, TX
MS in Artificial Intelligence — GPA: 3.7/4.0

Geethanjali college of Engineering and Technology — Hyderabad, India
B.Tech in Artificial Intelligence and Machine Learning — GPA: 3.1/4.0

Projects

Enterprise LLM Assistant with RAG Pipeline | *LangChain, Hugging Face, FAISS*

- Developed production-grade RAG system using Hugging Face Transformers and LangChain, integrating vector databases and similarity search for context-aware, grounded responses.
- Designed prompt engineering frameworks and evaluation pipelines to reduce hallucinations, benchmark semantic accuracy, and enforce responsible AI practices.

Automated ML Deployment Pipeline (MLOps) | *MLflow, Airflow, CI/CD*

- Built end-to-end CI/CD pipeline for systematized model retraining, versioning, and deployment, reducing time-to-production by 70% and enabling A/B testing of ML models.
- Deployed containerized inference services on cloud infrastructure with model monitoring and logging to ensure reliability and performance in production.

Predictive Analytics & Feature Engineering | *Machine Learning, Ensemble Models, Feature Engineering*

- Developed a machine learning ranking model using ensemble learning techniques and advanced feature engineering to improve recommendation quality and predictive performance.
- Achieved an NDCG score of 0.9274 through iterative feature selection, hyperparameter tuning, and rigorous model evaluation, enabling data-driven optimization and performance validation.

AI-Powered Visual Content Generator | *Stable Diffusion, CLIP, Computer Vision*

- Engineered a text-to-image generation pipeline using Stable Diffusion to create high-quality visual content from natural language prompts.
- Implemented CLIP-based text-image alignment scoring and optimized inference parameters, improving generation efficiency by 40% while maintaining output quality and semantic consistency.